

Alireza Falah

ROBOTICS ENGINEER (PERCEPTION & SIMULATION)

✉ al.r.falah@gmail.com | 🌐 alirezafalah.github.io/ | 🔗 linkedin.com/in/alireza-falah-5560b5116/



Fast Summary

Robotics Engineer bridging mechanical engineering foundations with modern software development. **(1) Simulation & Dynamics:** Developed high-fidelity, dynamics-aware autonomous racing environments in MuJoCo and Brax (Python/JAX) for the Dallara SF23, optimizing trajectories for the Autonomous Racing League competitions. **(2) Computer Vision Research:** Published researcher in Industry 4.0, designing end-to-end optical perception systems and researching fast, efficient 3D object reconstruction through monocular video and pose estimation. **(3) Robotics & HPC:** Experienced in optimizing complex autonomous stacks using C++ and ROS2, with a growing focus on parallel computing and acceleration using CUDA and oneAPI.

Work Experience

HUMDA Lab Nonprofit Ltd.

AI Researcher and Simulation Engineer

Budapest, Hungary

Feb 2024 - Present

- **Developed** and maintained a high-fidelity **dynamic-aware** racing simulation in **MuJoCo MJX** and **Brax** using Python/JAX. This simulation was used by the company to discover optimal, dynamically-feasible trajectories which was successfully used at the **A2RL 2025 Sim Challenge**.
- Adapted the company's autonomous software (developed in **C++**, **Python**, and **ROS2** to interface seamlessly with new simulation platforms, such as **automotive-grade simulator aiSim (Stellantis)** or **Aspire's Autonomia Autoverse**, requiring varied adapter configurations.
- Led the integration of **aiSim (Stellantis)** for complex autonomous driving simulations, engineering custom **virtual sensors for synthetic data generation**.
- Engineered the **data pre-processing pipeline** for raw LiDAR and Radar sensors, delivering clean, high-quality point clouds that were **used for training** the team's successful vehicle detection and tracking model, utilized in the **A2RL 2025 Multi Vehicle Test Sessions**.
- Maintained a complete **DevOps pipeline** using **GitHub Actions**, **Docker**, and **Azure VMs**. This pipeline automated the validation of the core autonomous software by running **Pytest-based quality gate scenarios** (e.g., tunnel passing, time trials) within a **containerized simulation**, ensuring continuous integration and robust performance for every commit.
- **Technical Skills:** C++, Python, ROS2, JAX, MuJoCo, Brax, Vehicle Dynamics, Docker, Pytest, DevOps, CI/CD, OpenCV, GitHub Actions, Azure, Git, Linux, CUDA, Synthetic Data Generation.

Deep Learning Center of Iran

Autonomous Vehicles Researcher

Tehran, Iran

Jan 2020 - Aug 2022

- Researched and implemented **high-performance perception modules** for autonomous racing vehicles, specifically optimizing **object detection and tracking** algorithms for real-time performance using **Python** and **ROS2**.
- Presented technical findings on perception architecture to engineering stakeholders, directly contributing to the organization's **autonomous technology roadmap**.
- **Tech Stack:** C++, Python, ROS2, OpenCV, Computer Vision, Object Detection (YOLO/SSD), Kalman Filters, Linux, Git.

Mapna Turbine Engineering and Manufacturing Company

Mechanical Engineer (Simulation & Analysis)

Karaj, Iran

Sep 2018 - Nov 2019

- Engineered Air Cooled Condensers (ACCs) by developing detailed Finite Element (FEA) models in Ansys, performing structural optimization to withstand thermal loads.
- Executed complex **numerical analysis**, solving governing **Partial Differential Equations (PDEs)** for nonlinear stress and stiff systems of **ODEs** for **transient thermal dynamics to validate simulation accuracy**.
- Modeled large-scale S-CO2 power cycles using Aspen Plus, analyzing system efficiency through rigorous process simulation.
- **Tech Stack:** Ansys (FEA), Matlab, Simulink, Aspen Plus, Numerical Analysis (PDEs/ODEs), Autodesk Inventor.

Education

Eötvös Loránd University (ELTE)

Szombathely, Hungary

M.S. in Mechanical Engineering (Specialization: Automation and Industry 4.0)

- **Stipendium Hungaricum Scholarship** recipient. **GPA:** 4.43/5.00
- **Thesis: Vision-Based Tool Monitoring.** Engineered an end-to-end **Computer Vision pipeline** using **OpenCV** and **Python**. Integrated industrial hardware (Basler) and implemented **hybrid signal processing** and **feature extraction** algorithms to quantify tool health. Resulted in the **ELTE-TCM-46k dataset** and publication in world Q1 journals.
- **Robotics Competition (TechTogether):** Secured 2nd Place solving industrial challenges for **Fanuc** and **Kuka** robots. Optimized robotic trajectories and integrated sensor feedback for pick-and-place tasks. Awarded special university scholarship for **consistent high performance** in competitive engineering events.
- Received training from industry experts at **Schaeffler** and **TDK**. Key courses: Computer Vision, Microcontroller Based Device Development, **Embedded Systems Programming (Assembly/Low-Level CPU-Specific)**, Industrial Communications, Digital Factory protocols.

Semnan University

Semnan, Iran

B.S. in Mechanical Engineering

- **Thesis:** Simulation utilizing **Finite Element Methods (FEM)** for deformable barrier testing. (Supervisor: Dr. Amir Najibi)
- Gained expertise in 3D Modeling, Solid Properties, Computational Fluid Dynamics, Finite Element Methods, Control and Automation, Heat Transfer, and Thermodynamics.

Projects

Personal R&D Project

Budapest, Hungary

Monocular 3D Reconstruction of Industrial Parts

Ongoing

- Developing a 3D reconstruction pipeline specifically for **reflective, textureless manufacturing parts** using monocular video sequences.
- Implementing robust **pose estimation** using **fiducial markers** to align video frames, generating **visual hulls**, and performing **feature mapping** to track object geometry across views.
- Targeting **metric-accurate 3D reconstruction** to generate precise digital CAD models, specifically solving scale ambiguity issues in single-camera setups.
- **Tech Stack:** Python, OpenCV (ArUco), 3D Geometry, oneAPI (Intel) / CUDA, Linear Algebra.

Industrial IoT: VFD Communication & SCADA Prototype

Szombathely, Hungary

Eötvös Loránd University

- Engineered a Python-based control interface for Variable Frequency Drives (VFDs), implementing the **Modbus RTU** protocol over RS485 for robust industrial communication.
- Developed a real-time dashboard (Tkinter) to visualize telemetry data and execute motor control commands, simulating an industrial SCADA environment.
- **Tech Stack:** Python, PyModbus, Industrial Comms (RS485), GUI Development.

Embedded Control: Self-Balancing Robot

Szombathely, Hungary

Eötvös Loránd University

- Prototyped an inverted pendulum system using an **Arduino** microcontroller, implementing a closed-loop **PID controller** to stabilize unstable non-linear dynamics.
- Executed the mechanical design (CAD/3D Printing) and integrated IMU sensors (Gyro/Accelerometer) for real-time state estimation.
- **Tech Stack:** C++, Embedded Systems, Control Theory (PID), Sensor Fusion.

Honors & Certifications

Abu Dhabi Autonomous Racing League (A2RL)

Yas Marina Circuit, Abu Dhabi, UAE

5th Place – International Autonomous Racing Competition

2024

- Achieved **5th Place** globally with the HUMDA Lab team, competing for a **\$2.5 Million prize pool** against elite research universities (e.g., TUM, UNIMORE).
- Executed critical **simulation-based validation** of vehicle behavior, ensuring control strategy safety and stability for the **Dallara Super Formula SF23** prior to track deployment.

Scientific Student Association (TDK)

Eötvös Loránd University

2nd Place (University Level) – National Finalist

2023

- Awarded **2nd Place** for Master's thesis in the Informatics section, securing qualification for the **National Scientific Students' Associations Conference (OTDK)**.
- Thesis selected for publication in the university's academic journal based on research excellence.

IBM Developer Skills Network

Online, Coursera

DevOps Essentials Professional Certificate

Issued 2026

- Covers Continuous Integration (CI/CD), Containerization (Docker), and Monitoring. ([View Badge](#))

Publications & Scientific Contributions

JOURNAL ARTICLES

Tool Condition Monitoring in CNC Systems: A Hybrid Computer Vision and Signal Processing Approach

Alireza Falah, Mátyás Andó, Bálint J. Szekeres

The International Journal of Advanced Manufacturing Technology (2025). 2025

Novel and Cost-Effective CNC Tool Condition Monitoring Through Image Processing Techniques

Alireza Falah, Mátyás Andó

Discover Applied Sciences (2025). 2025

CONFERENCE PROCEEDINGS

Autonomous Racing Competitions: A Review of Platforms, Software Development, and Sustainability Contributions

Zalán Demeter, Alireza Falah, Levente Puskás

Proceedings of the Conference on Sustainability (COS2024), 2024, Győr, Hungary

DATASETS

ELTE-TCM-46k: A CNC Tool Condition Image Dataset

Alireza Falah, Soninkhuu Baatarchuluun, Mátyás Andó, Béla Szekeres

Hugging Face repository, 2025

Skills

Languages & HPC

Python (JAX, PyTorch, Scikit-learn, NumPy), Modern C++, CUDA, oneAPI, Linux, Matlab, SQL

Robotics & Sim

ROS2 (Humble), MuJoCo, Brax, Nvidia Isaac Sim, aiSim, Vehicle Dynamics, Synthetic Data Gen, URDF/SDF

Computer Vision

OpenCV, 3D Reconstruction, Camera Calibration, Pose Estimation (ArUco), Visual Hull, Point Clouds (PCL)

Math & Physics

Finite Element Method (FEM), CFD, Numerical Methods (ODEs/PDEs), Monte Carlo, Rigid Body Dynamics

DevOps & Tools

Docker, GitHub Actions, CI/CD, Azure, Linux (Bash), Git, Foxglove, Microcontrollers (Arduino/STM32)

Soft Skills

Adaptability, Cross-Functional Collaboration, Communication, Creativity, Critical Thinking, Presentation Skills

References available upon request.